



Installation Conditions

Gas engine-generator sets
with **Series 400**, **Series 500**, and **Series 4000** engines

MS65032/08E

Product name	Engine model	Application *		Product name	Engine model	Application *
mtu 6R400 GS	E3066Z5 B3066Z8 E3066Lx E3066LHx E3066D3 E3066DH3 E3066D4 E3066DH4	3A		mtu 8V4000 GS	8V4000L32 8V4000L33 ** 8V4000L33 8V4000L64 8V4000L64FNER	3A
mtu 12V400 GS	E3042Z6 B3042Z7 E3042Lx E3042LHx E3042D3 E3042DH3 E3042D4 E3042DH4	3A		mtu 12V4000 GS	12V4000L32 12V4000L33 ** 12V4000L33 12V4000L64 12V4000L64FNER	3A
mtu 6R500 GS	E406-ct80 E406-ct70 E406-bt70 B406-ct135	3A		mtu 16V4000 GS	16V4000L32 16V4000L33 ** 16V4000L33 16V4000L64 16V4000L64FNER	3A
mtu 8V500 GS	E408-ct80 E408-ct70 E408-bt70 B408-ct135	3A		mtu 20V4000 GS	20V4000L32 20V4000L33 ** 20V4000L33 20V4000L64 20V4000L64FNER	3A
mtu 12V500 GS	E412-ct80 E412-ct70 E412-bt70 B412-ct135	3A				
* Continuous operation, unrestricted ** Reduced power						

Table 1: Applicability

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1 Preface

1.1 Preface

These Installation Conditions form the basic principles for application engineers, system/component designers and assembly personnel for planning or carrying out the installation of mtu components.

The Installation Conditions are applicable to all systems and gas engines of Series 400, Series 500, and Series 4000 supplied by Rolls-Royce Solutions, see (→ Business Portal).

The Installation Conditions do not relieve those responsible for the system from independently ensuring that the equipment is installed and checked in an appropriate manner.

The following accompanying documents must be complied with, in addition to the Installation Conditions:

- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating Instructions, see (→ Business Portal)
- Maintenance Schedule, see (→ Business Portal)
- Fluids and Lubricants Specifications, see (→ Business Portal)
- mtu standards

Refer to the specifications for the order concerned for information as to which documents apply.

Performing prescribed maintenance tasks properly and on schedule positively influences operational safety, reliability and service life. Ensure good accessibility for operation, maintenance and repair work when planning and installing the system.

Highlighted information

Important

This field contains product information which is important or useful for the user.

This information must not refer to hazards related to personal injury or material damage.

2 Product Information

2.1 Product safety

CE conformity of products from Rolls-Royce Solutions

The products from Rolls-Royce Solutions are designed and built in accordance with the Machinery Directive as well as other relevant EC directives. Rolls-Royce Solutions issues a “CE” declaration of conformity for complete machines according to the scope of delivery and provides such machines with a CE mark. For partly completed machines, a Declaration of Incorporation is issued.

In order to determine all hazards associated with a product, Rolls-Royce Solutions carries out a risk assessment. By implementing suitable protective measures, the hazards posed to human health and the environment are reduced or completely eliminated for all life stages in accordance with the state of the art. All important information on residual risks and on the safe and intended use of the machine can be found in the operating manual and other customer documents.

Integration of machinery / pressure devices / gas appliances

Rolls-Royce Solutions can issue declarations of conformity and incorporation only for scope of delivery. In order to determine possible safety-related linking between several machines / pressure devices / gas appliances, a higher-level risk assessment of the whole plant system is to be drawn up. The manufacturer of the complete plant is responsible for the preparation of the higher-order risk assessment, the implementation of the higher-order safety concept derived from it and the Declaration of Conformity with CE mark.

Depending on the plant system and the setup of the machine, some aspects may need to be considered:

- Internal and external explosions
- Fire
- Ventilation concept
- Heat utilization
- Biogas application
- Mechanical and electrical hazards
- Switching of the plant system, or a part thereof, to a safe state and protection against inadvertent start-up

The fulfillment of the conditions specified in this document is a prerequisite for the safe and smooth operation of the plant system.

Prior to the start of planning, an investigation into the requirements regarding official specifications has to be carried out and any required approvals obtained. The requirements for the building also apply to the container solutions, where appropriate.

Documents to be used:

- Technical description
- Technical data sheet
- Foundation data sheet
- P&I diagram
- Planning drawing
- Fluids and Lubricants Specifications
- Quality of safety components (→ Page 59)

3 Requirements for Building and Foundation

3.1 General requirements

The statics of the building must have a rating sufficient for the plant. If necessary, consult a structural engineer. Structural fire protection must be accomplished in accordance with the local laws and regulations. The foundation must be flat and such that it can safely support the weight of the engine-generator set and system components. The weight of the engine-generator set can be obtained from the technical data sheet. With regard to the vibrations caused by the engine-generator set, the foundation must be designed in accordance with the foundation data sheet.

Sufficient openings must be provided to install and remove components. The requirements of the louver dampers are described in the Ventilation chapter (→ Page 15). The building / container must have sufficient protection against lightning strokes. The plant must be integrated in the on-site lightning protection system. Specifications of insurance providers and authorities regarding lightning protection and fire protection must be researched (e.g. preparation of an integrated fire-protection concept, dividing into fire sections etc.).

Important

The technical data sheet and the foundation data sheet must be observed.

Important

To be observed in protection and danger zones (flooding areas):

When installing plants in protection and danger zones (flooding areas), general locally applicable laws, directives and standards must be observed.

4 Room Information

4.1 Premises – Requirements

To increase the level of work safety with regard to noise emission and vibration emission, it is recommended to install the engine-generator set and the switchgear in separate rooms, if possible. In so doing, the danger of the control cabinets overheating will be reduced. If a dedicated switchgear room cannot be provided, the installation site must comply with the requirements for the control cabinets delivered by Rolls-Royce Solutions.

Due to the noise emission, a sound-insulating enclosure is recommended for the engine-generator set. There must be sufficient lighting in the installation room and the room for the switchgear. The use of halogen spotlights with hot surfaces and high rates of heat loss is not permitted due to the risk of fire.

Emergency exits and escape routes must be well-lit and appropriately marked. Emergency exits and escape routes must not be blocked by cables or suchlike (pipes, hoses). The escape routes from other rooms must not pass through rooms for switchgear or the installation room. Escape doors must open in the direction of escape.

It must be possible to lock the installation room and the switchgear room, so that access by unauthorized persons can be prevented.

If the engine-generator set or the module is delivered by Rolls-Royce Solutions as a container solution, the container must be secured such that willful damage or unauthorized access by third parties can not occur.

4.2 Installation compartment – Requirements

For safe operation of the engine-generator set, the installation compartment must be suitably arranged. During the planning phase, you should already check whether compliance with further official requirements is necessary.

The installation compartment must be large enough to permit easy operation and maintenance of the module / engine-generator set as well as sufficient ventilation. The dimensions of the engine-generator set are shown in the planning drawing. The space required for movement during operation, maintenance and repair is also shown in the planning drawings. These dimensions required for movement must be observed in all circumstances. In addition to the dimensions of the engine-generator set and space required for movement, the space requirements of the peripheral connected systems (coolant circuits, exhaust system, oil system) must also be taken into consideration. Surfaces subject to heating must be insulated. The space required for insulation, the required minimum distance to walls and between the individual objects must be guaranteed. The erection of climbing aids must be possible. Ventilation ducts must be at least 0.7 m above the top edge of the mixture cooler.

Important

Comply with the planning drawing.

Lifting tools for maintenance work

Lifting equipment suitable for the heavy component parts of the engine and the exhaust system (turbocharger, mixture cooler, SCR element, etc.) must be provided in the installation compartment. For the maximum load, refer to the arrangement drawing of the engine-generator set.

The applicable laws and standards of the respective country of installation must be observed.

The installation of a crane rail or a mobile crane is necessary for the assembly work. A stop must be provided to limit the movement on removal beams and crane rails.

Important

Observe the planning drawing and the technical data sheet.

Fire protection requirements

Potential fire loads in the installation compartment must be minimized. In particular, no combustible material must be deposited or stored in the installation compartment. The minimum distance required for fire protection reasons between pipelines, other components and walls subject to heating must be observed. Wall ducts with pipelines and cables must be sealed off accordingly. Measures must be introduced for early detection of fires and smoldering fires, e.g. the installation of a fire detection system. Suitable extinguishing agents must be available (fire extinguisher, fire extinguishing system). No combustible material must be deposited or stored in the installation compartment.

The corresponding national legislation and local regulations on fire protection must be observed.

Explosion protection (OEG)

The Fluids and Lubricants Specifications must be observed.

The gas supply system before the inlet to the gas train must be designed to prevent the ingress of oxygen. If necessary, process-related access points such as condensate traps must be monitored.

According to the state of technology, suitable measures must be introduced because the possibility of an explosion in the exhaust system can not be excluded.

More detailed requirements are described in the Exhaust system chapter.

Exterior explosion protection (UEG)

The gas supply, including gas train, in the installation compartment must have a leak-tight design and must be checked on a regular basis for continuous leak-tightness. With circulation air systems, an exhaust air duct must be provided through which gas, in the event of a gas leakage, can be safely discharged. As a supplementary protective measure, the installation of a gas alarm system to be integrated in the plant control system is necessary. For this purpose, at least one gas warning sensor must be installed above the gas engine in direction exhaust air. Dead corners in the installation compartment without a flow should be avoided where possible because gas from leaks can accumulate there. If this is no longer possible for design-related reasons, gas warning sensors should be installed here. The room ventilation must be dimensioned such that, in the event of leakages on gas-carrying parts, a potentially explosive atmosphere can not be created. The gas-air mixture must be safely discharged directly to the atmosphere. For more detailed requirements of the ventilation system, refer to the Ventilation chapter.

Noise protection

Exact details on noise emission can be obtained from the technical data sheet. In accordance with these values, suitable measures must be taken, e.g. noise insulating paneling of the engine-generator set (acoustic enclosure) or acoustic boards in the installation compartment. The respective national legislation and local regulations must be observed.

Vibrations

The engine-generator set must be mounted on insulating boards to reduce the vibration transfer to the building and foundation. The exhaust gas aftertreatment system can be supported on insulating boards if necessary.

The length and number of the boards to be provided depends on the size and weight of the engine-generator set according to the planning drawing from Rolls-Royce Solutions. External vibrations must not act on the engine-generator set because this affects the natural frequency of the engine-generator set, which can be increased. This can result in failure or breakage of components.

Frost protection

The temperature in the installation compartment must not fall below 5 °C in any section. It may be necessary to install a heater.

Environmental protection

The installation compartment must comply with local water protection regulations (e.g. VAWS - Ordinance on Installations for the Handling of Substances Hazardous to Water and on Specialist Firms) and the WHG (Water Resources Act). If an extended oil system is used, the tank(s) must have a double-walled design and be equipped with a leakage indicator system. To reliably prevent the penetration of fluids and lubricants and water-polluting substances in the water system / groundwater, local regulations and requirements (oil separator, retaining device, leakage detection) must be observed.

Any other possible specifications (installation of an oil separator or a neutralization system for accumulated exhaust gas condensate) must be requested from the local authorities.

When installing or erecting plants in protection and danger zones, additional country-specific laws, directives and standards must be observed.

Ambient conditions (dust, moisture)

If control cabinets are located in the installation compartment, the ambient conditions must comply with the design of the control cabinets, (→ Page 12).

Protective equipotential bonding, protective conductor and grounding

For the selection and installation of electrical equipment, grounding arrangements and protective conductors, see DIN VDE 0100-540. Since the design of the protective equipotential bonding system and the protective conductor depends on the local mains type, details can not be discussed here. Only generally applicable information independent of the form of power grid can be provided at this point.

Protective equipotential bonding

Apart from the engine-generator set and heat module frame, all other metallic, electrically conductive components must be included in the local protective equipotential bonding system. This includes the gas train, heating water pipes, the gas line, the exhaust system and the main components of the exhaust gas after-treatment system, cable racks and control cabinets.

There are several connecting points for the protective equipotential bonding are ready for use at the engine-generator set frame. Protective equipotential bonding extending from the engine-generator frame to the generator is provided through flexible connecting lines.

Protective conductor

The protective conductor must be connected in every case to the housing of the generator. Grounding of the generator neutral point depends on plant-specific factors. The local regulations of the power supply company or the safety regulations must be observed to ensure that grounding is carried out correctly.

ZEP - Central grounding point

State of the art is the version with a central grounding point. This reliably prevents stray currents. If a central grounding point is used, there is only one connection between PEN (protective conductor with neutral conductor function) and ground. The complete run of the PEN conductor must be insulated. For networks with decentralized power supply, a further central ground point is possible. The two areas with central grounding points must not be operated in parallel. A central grounding point must be provided in the installation compartment.

Lightning protection and surge arrester

In low voltage distribution systems and consumer systems, overvoltage can occur caused by lightning, switching operations in the higher-order power grid or short circuits. These high electric voltages can damage or even destroy electrical and electronic systems and devices. Overvoltage protectors (Surge Protective Device, SPD) have the task of discharging these overvoltages and high-energy lightning currents in direction ground and, in the downstream installation, limiting the voltage that is present to a maximum permissible voltage.

The protective systems that are used have a multi-layer structure and the extent to which they are necessary is typically based on the local situation.

- Overvoltage protectors of type 1 (SPD type 1), so-called lightning stroke current arresters are necessary if high lightning currents can couple via the ground or the arresting devices into the equipotential bonding conductor of the low-voltage system.
- Overvoltage protectors of type 2 (SPD type 2), so-called surge arresters, are used as the second protection level after the lightning stroke current arresters and for limitation of lightning surges from remote lightning strikes or from switching operation overvoltages.
- Overvoltage protectors of type 3 (SPD type 3) are used near to the device to be protected (e.g. cable channel, sockets, directly before a computer).

Support for the concrete implementation of lightning and overvoltage protection is provided in, e.g. the standards EN 62305, IEC 60364-4-44 and IEC 60364-5-53.

As the necessity of lightning and overvoltage protection depends on the local situation and, in many projects, the upstream installation of the customer's installations typically already uses protective devices, the standard version of engine-generator sets from Rolls-Royce Solutions does not have its own lightning and overvoltage protection installed. If necessary, consult your contact person at Rolls-Royce Solutions.